RAJALAKSHMI ENGINEERING COLLEGE

V Kanyalakshme 251801130-2025-AIDS

V2

Started on	Thursday, 8 October 2026, 10:25 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:29 AM
Time taken	3 mins 43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, k, i = 0, j = 1;
    if (scanf("%d", &n) != 1) return 0;

    int A[n];
    for (int m = 0; m < n; m++) scanf("%d", &A[m]);
    scanf("%d", &k);

    while (i < n && j < n) {
        int diff = A[j] - A[i];
        if (i != j && diff == k) return printf("1\n"), 0;
        if (diff < k) j++;
        else if (++i == j) j++;
    }
    return printf("0\n"), 0;
}

```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

V Kanyalakshme 251801130-2025-AIDS

V2

Started on	Thursday, 24 September 2026, 10:24 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:48 AM
Time taken	23 mins 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main(){
    int n,i,j;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    for(i=0;i<n;i++)
        for(j=i+1;j<n;j++)
            if(a[i]==a[j]){
                printf("%d",a[i]);
                return 0;
            }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

V Kanyalakshme 251801130-2025-AIDS

V2

Started on	Thursday, 24 September 2026, 10:56 AM
State	Finished
Completed on	Thursday, 24 September 2026, 11:30 AM
Time taken	34 mins 43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int n,a[100],i,j;
    scanf("%d",&n);
    for (i=0;i<n;i++)
        scanf("%d", &a[i]);
    for (i=0;i<n;i++)
    for (j=i+1;j<n;j++)
            if (a[i]==a[j]){
                printf("%d",a[i]);
                return 0;
            }
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

V Kanyalakshme 251801130-2025-AIDS

V2

Started on	Thursday, 24 September 2026, 11:31 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:55 AM
Time taken	13 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

void solve() {
    int n1, n2, i = 0, j = 0;
    if (scanf("%d", &n1) != 1) return;
    int a[n1]; for (int k = 0; k < n1; k++) scanf("%d", &a[k]);
    if (scanf("%d", &n2) != 1) return;
    int b[n2]; for (int k = 0; k < n2; k++) scanf("%d", &b[k]);

    while (i < n1 && j < n2) {
        if (a[i] == b[j]) {
            if (i == 0 || a[i] != a[i - 1]) printf("%d ", a[i]);
            i++; j++;
        }
```

```
        else if (a[i] < b[j]) i++;
        else j++;
    }
    printf("\n");
}

int main() {
    int t;
    if (scanf("%d", &t) == 1) while (t--) solve();
    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

V Kanyalakshme 251801130-2025-AIDS V2

Started on	Thursday, 8 October 2026, 10:15 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:18 AM
Time taken	3 mins 9 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1, n2;


    if (scanf("%d", &n1) != 1) return;
    int *a = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &a[i]);
    }


    if (scanf("%d", &n2) != 1) {
        free(a);
        return;
    }
    int *b = (int *)malloc(n2 * sizeof(int));
```

```
    for (int i = 0; i < n2; i++) {
        scanf("%d", &b[i]);
    }


    int i = 0, j = 0;
    int first = 1;

    while (i < n1 && j < n2) {
        if (a[i] < b[j]) {
            i++;
        } else if (a[i] > b[j]) {
            j++;
        } else {

            if (!first) {
                printf(" ");
            }
            printf("%d", a[i]);
            first = 0;
            i++;
            j++;
        }
    }
    printf("\n");


    free(a);
    free(b);
}

int main() {
    int t;
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

RAJALAKSHMI ENGINEERING COLLEGE

V Kanyalakshme 251801130-2025-AIDS V2

Started on	Thursday, 8 October 2026, 10:20 AM
State	Finished
Completed on	Thursday, 8 October 2026, 10:25 AM
Time taken	5 mins 31 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int has_pair_with_diff(int A[], int n, int k) {
    int i = 0;
    int j = 1;


    while (i < n && j < n) {
        int diff = A[j] - A[i];

        if (diff == k && i != j) {
            return 1;
        } else if (diff < k) {
            j++;
        } else {
            i++;
            if (i == j) {
                j++;
            }
        }
    }
    return 0;
}

int main() {
    int n, k;


    if (scanf("%d", &n) != 1) return 0;


    int *A = (int *)malloc(n * sizeof(int));
    if (A == NULL) return 1;


    for (int i = 0; i < n; i++) {
        if (scanf("%d", &A[i]) != 1) {
```

```
            free(A);
            return 0;
        }
    }


    if (scanf("%d", &k) != 1) {
        free(A);
        return 0;
    }


    printf("%d\n", has_pair_with_diff(A, n, k));

```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course